A comment on 'Growth and Inequality in Public Good Provision': Testing the Robustness and Generalizability of Dynamic Public Good Games

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Online Appendix

1. Online Feasibility

Throughout this section, I do not consider GMTVs' data and pool data I gathered from the citizens as well as the students from Hamburg (N = 151).

How did the participants, who have never participated in an online group experiment before, cope with the situation? Moreover, did participants understand the unfamiliar setting they found themselves in? While the answer to the former question requires more thought, the answer to the latter simply is *yes*: 67 out of 116 answered with *"yes"* when I asked them. Another 44 answered with *"rather yes"* while nobody indicated that he or she did not understand the situation at all. Some behavioral data are supporting this finding: The user interface offered a popup to review instructions or contact information. I tracked both and find that none of the participants ever opened these popups even though they were clearly visible in the decision screens' header and introduced in the instructions. To further analyze how participants coped with the situation, I consider three additional metrics: selection into the experiment, attrition as well as the time spent on each page.

I first comment on the selection into the experiment: It was difficult to recruit the sample. The panel counted 1.209 non-students of which I was able to recruit 130 participants who finished the experiment—even though I varied the weekdays and timing of the sessions (which were conducted during a nationwide lockdown under mandatory home office regime). For this reason, I also recruited students in the last session which explains the relatively large number of showups in Table 1. Although I intended to refrain from the recruitment of students initially, this particular sub-sample enabled me to investigate the generalizability of my results as I discuss in Section 3.3.

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Table 1: The Experimental Sessions' Meta Data

Session Code	Date	Time	Showups	Dropouts	Residuals	Participants	Observations
jyf8xd0s	2021-07-01	15:00	35	4	3	28	7
vggk2gh1	2021-07-03	13:00	20	8	0	12	3
8gi7c8xg	2021-07-09	13:00	21	5	4	12	3
d6jrsxnr	2021-07-23	14:00	75	8	3	64	16

Turning to the time spent on each page, I focus on the decision times in the dynamic public goods game as Anderhub et al. (2001) did. How many seconds did the participants need to decide in each period of the game? Not too many. Figure 1 illustrates an plausible pattern: The first decision took about 22 seconds. The second decision—where participants first learned about the other group members' previous decisions—took longer (about 33 seconds). Subsequently, decision times first declined and stabilized at 19 seconds. Importantly, decision times were so short that crosstalk, that is, communication through private channels—a common concern[1] in online experiments—was unlikely, especially because it would require the identification of other group members.[2]

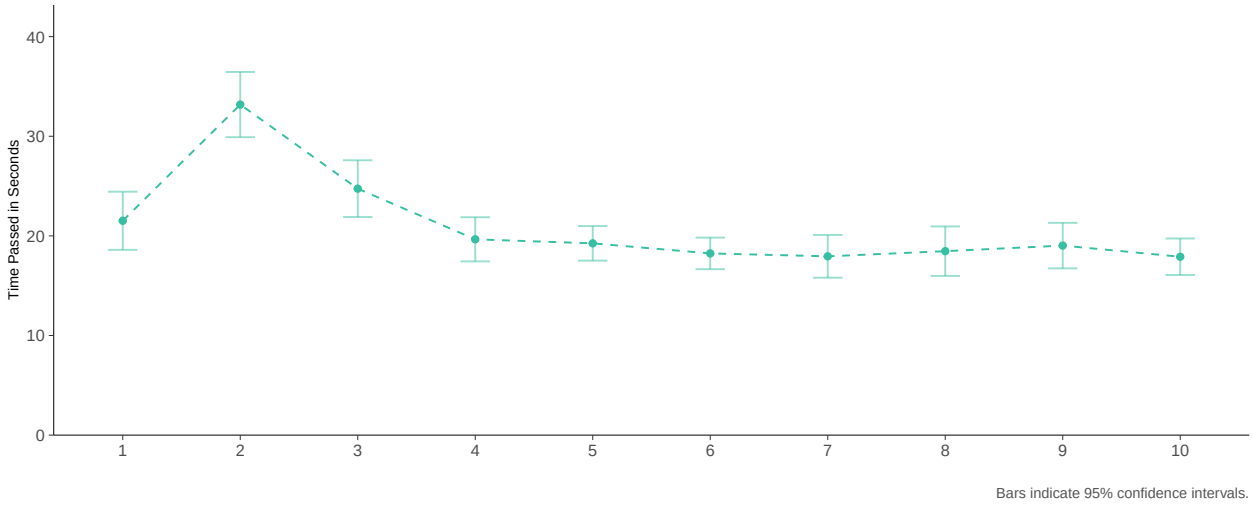


Figure 1: Average Time Spent for Each Contribution per Period

Considering attrition, I find that it did not affect the interactive experiment at all. To elaborate, I differentiate between dropouts and residuals: Participants who could not be matched to other group members are called residuals. Participants who intentionally left the experiment are called dropouts. Residuals did

[1] See, for instance, the discussion section in Arechar et al. (2018, p. 119).

[2] There were only 9 participants (from all four sessions) who needed more than 60 seconds to make the second decision.

not participate in the experiment *by design*. Dropouts did not participate in the experiment *by choice*. Out of 151 people who showed up, I count 10 residuals and 25 dropouts. All of the residuals waited to be matched to a group unsuccessfully before they got paid one Euro for their patience. In contrast, all of the dropouts left while reading the instructions and before being matched to other group members. Moreover, they received no payment at all. Hence, attrition was no concern considering the dynamic public goods game or the expenses.

2. Comprehension

The participants in this study were grouped based on the speed at which they answered comprehension questions, with higher group IDs indicating that those participants completed the questions later. This could introduce a potential bias in how participants were grouped in terms of their comprehension levels. Specifically, participants who were quicker to answer comprehension questions may have had a better understanding of the task, and thus their comprehension scores could be systematically higher than those who took longer to answer. This grouping by answering speed might inadvertently align with comprehension ability, potentially leading to confounding effects in the results.

To test this potential bias, I estimate a linear mixed-effects model specified as:

$$y_{ij} = \beta_0 + \beta_1 \text{GroupID}_{ij} + u_j + \epsilon_{ij}$$

where y_{ij} represents the comprehension score for participant i in session j, GroupIDij indicates their grouping order, u_j captures session-specific random effects, and ϵ_{ij} is the error term. The model tests whether faster completion of comprehension questions predicts better understanding while accounting for potential session-level variations through random intercepts.

The results presented in Table 2 suggest that group ID does not significantly predict comprehension. The coefficient for group ID is small and insignificant (β = -0.02, se = 0.01), indicating that the order in which participants were grouped based on their response speed does not have a significant effect on their comprehension scores in this sample.

Table 2: Linear Mixed-Effects Model for Comprehension, Accounting for Group ID and Session Effects

	Mixed Effects Model
	Comprhension
Group ID	−0.02 (0.01)
Constant	2.65*** (0.16)
Observations	116
Log Likelihood	−101.69
Akaike Inf. Crit.	211.38
Bayesian Inf. Crit.	222.39
Note:	*p<0.1; **p<0.05; ***p<0.01

References

Anderhub, V., Müller, R., Schmidt, C., 2001. Design and evaluation of an economic experiment via the internet. Journal of Economic Behavior and Organization 46, 227–247. URL: https://www.sciencedirect.com/science/article/pii/S0167268101001950, doi:https://doi.org/10.1016/S0167-2681(01)00195-0.

Arechar, A.A., Gächter, S., Molleman, L., 2018. Conducting interactive experiments online. Experimental Economics 21, 99–131. URL: https://doi.org/10.1007/s10683-017-9527-2, doi:10.1007/s10683-017-9527-2.